

SPN 521160/FMI 4 – SSAM02T

Backup Lamp 1 – Voltage Below Normal

Table 1, SPN 521160/FMI 4

Description	Backup Lamp 1 voltage is below normal or is shorted to a low source.
Monitored Parameter	N/A
Enable Conditions	The SSAM detects voltage below normal on connector 6, pin 25.
Monitored Sequence	Continuous
Execution Frequency	Always
Typical Duration	2 seconds
HMI Indication	None
Symptom	Tractor backup lamps are inoperative.
Verification	Tractor backup lamps are operational.

Table 1, SPN 521160/FMI 4

Table 2, Wiring

Description	ECU	Connector/Pin	Component	Connector/Pin	Module	Circuit
BACKUP LAMP 1	SSAM02T	6/25	TAIL LAMP, LEFT	BACKUP/A	–	–
TAIL LAMP, LEFT	SSAM02T	5/8	TAIL LAMP, LEFT	TAIL/B	–	–
STOP/TURN LAMP, LEFT	SSAM02T	8/18	TAIL LAMP, LEFT	STOP-TURN/C	–	–
EXTERIOR LIGHTING GND P #3	SSAM02T	11/2	TAIL LAMP, LEFT	LEFT GND/E	–	–
EXTERIOR LIGHTING GND P #3	SSAM02T	11/2	TAIL LAMP, RIGHT	RIGHT GND/E	–	–
STOP/TURN LAMP, RIGHT	SSAM02T	6/19	TAIL LAMP, RIGHT	STOP-TURN/C	–	–

TAIL LAMP, RIGHT	SSAM02T	8/8	TAIL LAMP, RIGHT	TAIL/B	–	–
BACKUP LAMP 1	SSAM02T	6/25	TAIL LAMP, RIGHT	BACKUP/A	–	–

Table 2, Wiring

Fault Code Troubleshooting

1.

☐ With battery switch in the ON position, turn the key ON and connect to DiagnosticLink®. Are faults for low voltage being reported by other modules

a.

☐ **YES** → Repair as needed. Verify the repair.

b.

☐ **NO** → Go to step 2.
2.

☐ Verify that the vehicle is correctly parameterized. Is the vehicle correctly parameterized

a.

☐ **YES** → Go to step 3.

b.

☐ **NO** → Correct the vehicle parameters, then verify the repair.
3.

☐ Check voltage between cavity A at the tractor tail lamp plug and battery positive (+). Is voltage present

a.

☐ **YES** → Go to step 4.

b.

☐ **NO** → Clear the fault, then verify the repair.
4.

☐ Disconnect Single Signal and Actuation Module (SSAM) connector 6. Check for voltage between pin 25 (harness side) and battery positive (+). Is voltage present

a.

☐ **YES** → Repair the short to ground in the wiring harness, then verify the repair.

b.

☐ **NO** → Go to step 5.
5.

☐ Perform a fixed load voltage drop test on the circuit. See [Fixed-Load Voltage Drop Test](#). Does the circuit pass the test?

a.

☐ **YES** → Install a test SSAM and attempt to clear the fault. If the fault clears, the vehicle SSAM may be faulty.

▪ [Replacement of the Single SAM](#) in the *New Cascadia Workshop Manual*

▪ [Replacement of the Single SAM](#) in the *47X & 49X Workshop Manual*

▪ [Replacement of the ASAM](#) in the *57X Workshop Manual*

b.

☐ **NO** → Repair the circuit, then verify the repair.
- NOTE:
- Component Code / VMRS = 034-003-074 – LAMP ASSEMBLY, BACKUP

Component Code / VMRS = 034-002-035 — WIRING HARNESS, BACKUP LAMP

Component Code / VMRS = 037-001-012 — MODULE, SIGNAL ACTIVATION (SAM), SINGLE

Note:
F07,F15,F16,F18,F25,F26,F35,F36,F41,F42,F43,F44,W12,W13,W14,W15,W16,W17,W18
SPN521160/FMI4
SSAM02T
TROUBLESHOOT
034-002-035,034-003-074,037-001-012

Parent topic: [SSAM02T \(SA 33\)](#)
Document Number: 0000085675
Manual Publication Date: 2024-06-05
Topic Publication Date: 2025-06-30

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